Practice Exam - 1

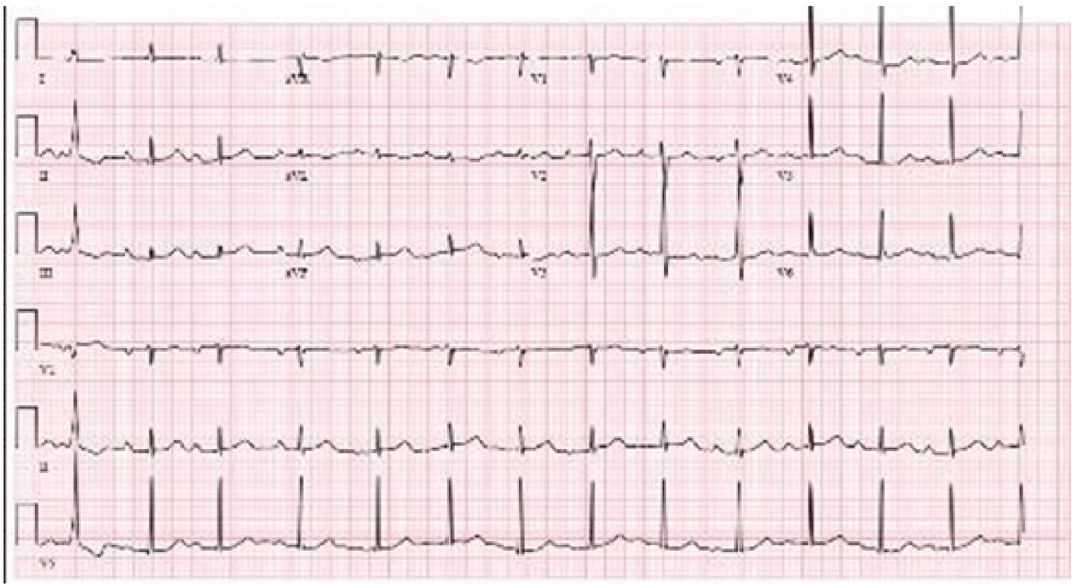
Question 70 of 117

A 65-year old with ischemic cardiomyopathy presents for a follow-up outpatient visit for progressive shortness of breath. The patient has been hospitalized twice in the last year for decompensated heart failure despite adherence with diet and medications. Due to symptomatic hypotension the dose of sacubitril- valsartan was decreased from 97-103 mg twice daily to 24-26 twice daily.

After discharge, a cardiopulmonary exercise test revealed a peak exercise VO2 of 12 ml/min/kg with an RER of 1.2.

12-lead electrocardiogram is shown below. Echocardiogram shows a left ventricular end-diastolic dimension of 8.0 cm and a left ventricular ejection fraction of 15% with severe mitral regurgitation. Current medications include sacubitril-valsartan 24-26 mg BID, metoprolol succinate 25 mg daily, spironolactone, 12.5 mg daily and torsemide 80 mg daily.

Which is the next best step in management?



ECG

A. Refer to transcatheter mitral valve repair

B. Refer for implantation of pulmonary artery pressure sensor

C. Refer for evaluation for heart transplantation

D. Refer for cardiac resynchronization therapy

E. Considered advanced heart failure medical therapy with addition of inotrope (e.g., intermitted miirinone)

Commentary References

Testing Point: The Board-certified AHFTC specialist should recognize when to refer a patient with end-stage heart failure for evaluation for heart transplantation.

Correct Answer: C. Refer for evaluation for heart transplantation.

Rationale: The patient has advanced heart failure as indicated by repeated hospitalizations for heart failure, intolerance to neurohormonal blockade due to hypotension with persistent symptoms. His cardiopulmonary exercise test (CPET) demonstrates severely impaired functional capacity in the setting of maximal volitional effort with predicted improvement in overall survival with heart transplant.

Answer B: Although this patient may benefit from an implantable pulmonary artery pressure sensor given repeated hospitalizations, he has evidence of progressive disease and should be evaluated for heart transplantation in a timely manner.

Answer A: With respect to transcatheter mitral valve repair, the patient has severe functional mitral regurgitation, but given a severely dilated left ventricle with an LVEDD of 8.0 cm, is unlikely to benefit from this intervention.

Answer D: The patient does not have a sufficient prolongation of the QRS to warrant consideration of cardiac resynchronization therapy.

Answer E: Continuous inotropic therapy may be used as temporary support or as bridge-to-transplant therapy in select patients with advanced heart failure. However, while inotropes may provide short-term hemodynamic or symptomatic improvement, they do not replace evaluation for definitive advanced therapies. In this patient with progressive Stage D heart failure and severely impaired functional capacity, referral for heart transplantation should not be delayed.